

DecodeLabs Internship

Project 2 Report

Project Title: Supervised Learning - Fraud Detection Pipeline

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Objective: Build a supervised machine learning pipeline capable of identifying suspicious transactions using classification algorithms and proper evaluation metrics.

1. Introduction

Fraud detection is one of the most important applications of supervised machine learning. The aim of this project is to build a complete fraud detection workflow that preprocesses transaction data, handles class imbalance, trains multiple classification models, and evaluates them using reliable metrics. Because the supplied dataset did not contain an actual fraud label, a heuristic target variable (IsFraud) was engineered from high-risk transaction characteristics for demonstration purposes.

2. Dataset Description

The dataset contains e-commerce transaction records including Quantity, Unit Price, Total Price, Payment Method, Order Status, Coupon Code, Referral Source and transaction Date. Additional temporal features such as Year, Month and Day of Week were extracted from the Date column.

3. Methodology

Step	Description
Data Loading	Imported CSV dataset using Pandas.
Feature Engineering	Generated Year, Month and DayOfWeek from Date.
Fraud Label	Created heuristic IsFraud target using transaction thresholds.
Preprocessing	Applied StandardScaler and OneHotEncoder.
Class Imbalance	Handled minority class using SMOTE.
Model Training	Trained Logistic Regression and Random Forest.
Hyperparameter Tuning	Used GridSearchCV to optimize model parameters.
Evaluation	Compared Accuracy, Precision, Recall, F1-Score and ROC-AUC.
Result Export	Saved model comparison and labelled dataset.

4. Machine Learning Pipeline

The workflow consisted of train-test splitting, preprocessing with a ColumnTransformer, SMOTE oversampling to balance the classes, and training two supervised learning algorithms: Logistic Regression and Random Forest. Hyperparameter tuning was performed using GridSearchCV. The final models were evaluated on unseen test data.

5. Evaluation Metrics

The models were evaluated using Accuracy, Precision, Recall, F1-Score and ROC-AUC. Confusion matrices and ROC curves were generated to compare classification performance. The model with the highest ROC-AUC score was selected as the final model.

6. Tools & Libraries

Python, Pandas, NumPy, Scikit-learn, Imbalanced-learn (SMOTE), Matplotlib and Jupyter Notebook.

7. Conclusion

This project demonstrates an end-to-end supervised learning workflow including preprocessing, feature engineering, handling imbalanced data with SMOTE, model training, hyperparameter tuning and evaluation. Although the fraud labels were engineered due to the absence of a true fraud target in the dataset, the notebook successfully demonstrates the techniques required for a fraud detection pipeline.